

CSCN8020 Assignment 2

Taxi Gymnasium Environment with Q-Learning

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Course	CSCN8020 - Reinforcement Learning Programming

Project summary

This project applies tabular Q-learning to Gymnasium's Taxi-v3 environment, where an agent learns to navigate a 5 x 5 grid, pick up a passenger, and make a legal delivery. The solution separates environment management, agent behaviour, training, metrics, plotting, experiments, and logging into meaningful classes. It evaluates the required baseline ($\alpha = 0.1$, $\epsilon = 0.1$, $\gamma = 0.9$), controlled learning-rate and exploration tests, and a final evidence-based configuration. The notebook tracks returns, steps, success rates, and illegal actions, produces plots, evaluates the learned policy greedily, and connects the implementation to Q-learning pseudocode, temporal-difference updates, and the course's Monte Carlo variance discussion.

GitHub repository

Repository link	https://github.com/chooksemmanuel/CSCN8020_Assignment2
Cloneable .git URL	https://github.com/chooksemmanuel/CSCN8020_Assignment2.git

Submission note

Brightspace submission: upload this one-page PDF and provide the repository link plus the cloneable .git URL shown above. The GitHub repository contains the executed notebook, plots, log file, experiment summary, README, requirements.txt, and .gitignore.